

A Calibrated, Preprocessing-Free Rank Test for Deterministic Structure in Time Series: Operation Order, Measured Surrogate Limits, and an Open Benchmark

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Abstract

Version 1 of this work proposed DDBM, a screening test that quantizes a rank-normalized scalar series onto an integer lattice, builds a modular phase from a cubic kernel, and reads departures from uniformity as evidence of deterministic structure. This version keeps the goal, discards most of the construction, and delivers what survives as a validated procedure.

What we removed. The cubic kernel turns out to be one member of a cyclotomic family: $S_p(N) = (N+1)^p - N^p$ factors into homogeneous cyclotomic polynomials at consecutive integers, so $(N+1)/N$ is an exact p -th root of unity modulo S_p and the phase lattice has denominator $\Phi_n(1)$. The identity is correct and, to our knowledge, previously unstated in this context, but it buys nothing: because Ξ is a deterministic function of the pair $(N_t, \Delta N_t)$, the phase map is a lossy projection, and a direct occupancy count on that pair scores 35/40 against 34/40 for the phases on our labeled benchmark. The modular arithmetic is discarded.

What we found. The order of operations decides whether the test is valid at all. Detrending, AR(1) prewhitening and volatility standardization applied *before* rank normalization destroy distribution-freeness: empirical size at nominal 0.05 reaches 0.19 for a lognormal marginal and 0.88 for a Cauchy marginal. Working on ranks alone restores exact distribution-freeness, with a one-line proof, and returns nominal size across six marginals. We also audit the surrogate stage, which is where every verdict ultimately rests: it rejects a static monotone transform of a linear process—inside its own null—in 25% of replicates, and heteroscedastic processes in 50–75%. That measurement invalidates the financial conclusion of v1 and of our own earlier draft.

What we recommend. A preprocessing-free, rank-based test with exact Monte-Carlo calibration of the full scan, IAAFT confirmation, and documented confound controls. Under matched conditions the best discriminating statistic is the classical missing-ordinal-pattern count with the pattern length scaled to the sample (37/40); pair occupancy follows at 35/40. We claim no new detection principle: the mechanism is the forbidden/missing-pattern paradigm and coarse-grained transition statistics. The contribution is the calibrated decision layer, the measured limits, and an open benchmark of 165 series whose synthetic labels are computed from Lyapunov exponents rather than asserted.

1 What this version claims, and what it withdraws

Version 1 claimed a new detector: a modular-arithmetic phase statistic reaching 92.5% accuracy on a 40-system benchmark, “without phase-space reconstruction or training data”. Three of those claims do not survive scrutiny.

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The method is not embedding-free. The phase Ξ_t is a function of the pair (N_t, N_{t+1}) , i.e. of a two-dimensional delay vector with $m = 2$, $\tau = 1$, quantized at resolution $1/K$. Scanning K is a scan over box scale, and the count of realized pairs is a single-scale box count in that embedding. Embedding parameters were not eliminated; they were fixed and hidden. The statement in v1 is withdrawn.

The stated mechanism contradicts the preprocessing. Version 1 attributed the signal to a singular marginal measure occupying few of K bins. Rank normalization removes exactly that: for a continuous marginal and $n \gg K$, every bin receives about n/K points regardless of how singular the underlying measure is. Whatever the test detects, it is not marginal sparsity; it is the geometry of *transitions*. This reinterpretation is forced, and it is what the present version analyses.

The p -values were not valid. Phases from overlapping pairs are serially dependent, while the two-sample Kolmogorov–Smirnov test used to score them assumes independence; a Bonferroni correction over the scanned K does not repair a miscalibrated individual test. Section 3 shows the defect is worse than a dependence artifact: with the published order of operations the procedure is not even distribution-free.

We also state plainly what is *not* new. Discriminating determinism from noise by the occupancy of symbol transitions is classical: symbolic dynamics and forbidden words [1], the Bandt–Pompe ordinal symbolization [2], forbidden and missing ordinal patterns [3, 4], ordinal transition networks [5], and direct determinism tests [6, 7]. The confound analysed in Section 6 — smooth stochastic processes imitating low-dimensional determinism — has been known since [8, 9]. Nothing in this paper is a new detection principle.

2 The cyclotomic identity, and why it does not help

2.1 The identity

With $S_p(N) = (N+1)^p - N^p$ we have, by construction, $(N+1)^p \equiv N^p \pmod{S_p(N)}$.

Lemma 1 (order of the consecutive ratio). *Let $N \geq 1$ and $p \geq 2$. Then $\gcd(N, S_p(N)) = 1$, and $\rho = (N+1)N^{-1} \pmod{S_p(N)}$ has multiplicative order exactly p .*

Proof. Expanding, $S_p(N) = \sum_{j < p} \binom{p}{j} N^j \equiv 1 \pmod{N}$, so N is invertible. From $(N+1)^p \equiv N^p$ we get $\rho^p \equiv 1$, so the order divides p . If the order were $d < p$ with $d \mid p$, then $S_p(N)$ would divide $(N+1)^d - N^d$, which is a positive integer strictly smaller than $S_p(N)$ for $N \geq 1$ and $d < p$ — impossible. Hence the order is p . $\square$

The degenerate cell $N = 0$ must be excluded: $S_p(0) = 1$, the residue ring is trivial and N^{-1} does not exist. Implementations should either drop that cell or map it to a fixed phase; v1 did neither.

Expanding the difference of powers gives the homogeneous cyclotomic factorization

$$S_p(N) = \sum_{j=0}^{p-1} (N+1)^j N^{p-1-j} = \prod_{d \mid p, d > 1} \Phi_d(N+1, N), \quad (1)$$

verified symbolically for $p = 2, \dots, 6$; e.g. $S_4 = (2N+1)(2N^2+2N+1) = \Phi_2\Phi_4$ and $S_6 = (2N+1)(N^2+N+1)(3N^2+3N+1) = \Phi_2\Phi_3\Phi_6$. Because the arguments are coprime, the classical primitive-divisor theory of Bang and Zsigmondy [10] applies: every prime divisor q of $\Phi_p(a, b)$ satisfies $q \equiv 1 \pmod{p}$ except possibly

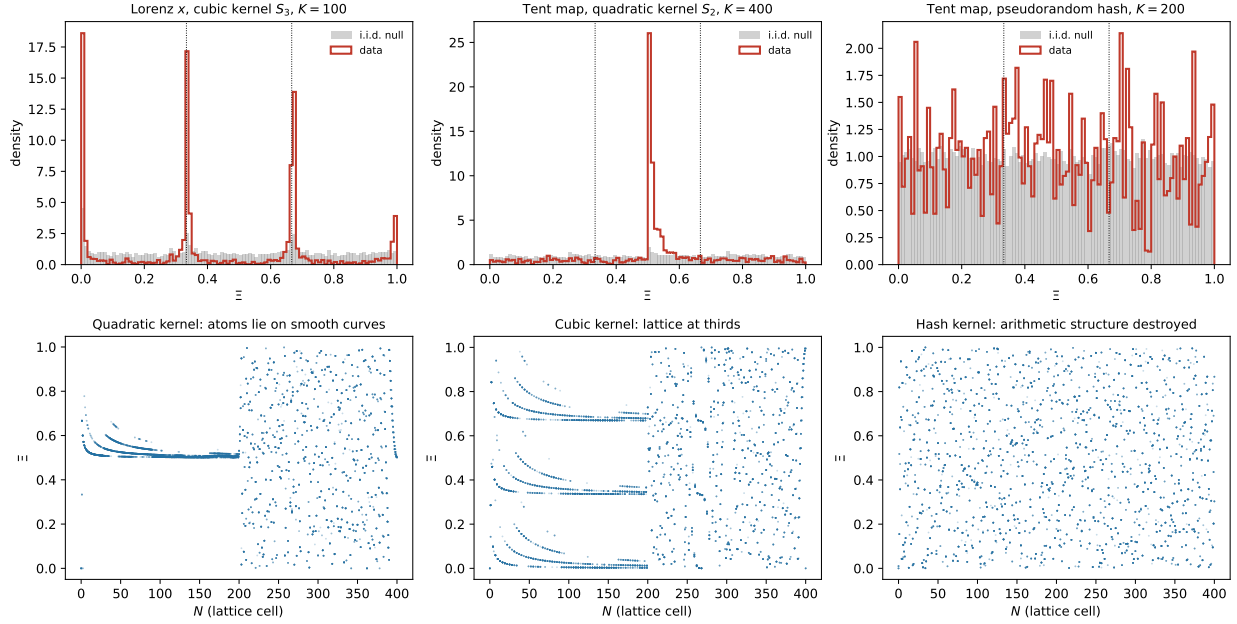


Figure 1: Phase histograms against the i.i.d. null (top) and the atoms Ξ against the lattice cell N (bottom). Lorenz under the cubic kernel concentrates at $\{0, \frac{1}{3}, \frac{2}{3}\}$; the tent map under the quadratic kernel shows the exact resonance at $\frac{1}{2}$; a pseudorandom hash on the same data retains the atom count but destroys the arithmetic positions.

$q = p$. We verified no violations for $p \in \{3, 5, 7\}$, $N \in [2, 200]$. Consequently p divides $|(\mathbb{Z}/S_p)^\times|$ and an element of order p must exist — Lemma 1 is not an accident of small cases.

For $p = 3$ the identity $3N^3 = S_3(N)(N-1) + (2N+1)$ places the phases at $\Xi(N, 0) = k/3 + O(1/N)$ with k determined by $N \bmod 3$; the median distance to $\{0, \frac{1}{3}, \frac{2}{3}\}$ is 2.2×10^{-4} over $N \in [2, 2000]$. More generally the lattice denominator equals $\Phi_n(1)$, which is p for $n = p^k$ and 1 otherwise; measured denominators match for $n = 2, \dots, 12$.

We emphasise the scope: this is proved for $\Delta N = 0$, and for the special branch of the tent map where $\Delta N = N$ yields the exact resonance $\Xi = (N+1)/(2N+1) \rightarrow \frac{1}{2}$. That an *arbitrary* deterministic trajectory concentrates on the lattice more strongly than a stochastic one is an empirical observation (Fig. 1), not a theorem.

2.2 Why the arithmetic is nevertheless discarded

Ξ is a deterministic function of $(N_t, \Delta N_t)$. Any test on Ξ is therefore majorized by a test on the pair itself: the phase map can only lose information. Empirically it does. Under a matched protocol (Section 4, preprocessing-free), a plain count of distinct realized pairs scores 35/40 against 34/40 for the calibrated cyclotomic phase scan, and the classical ordinal baseline scores 37/40 (Table 3).

An earlier draft of this version reported that cyclotomic kernels give “5–8 $\times$ sharper” z -scores than power kernels. That statement is withdrawn: the quoted values ($z \approx 750$ against a null simulated with a few hundred replicates) lie far outside the support of the simulated null and cannot be read as sensitivity. At matched nominal size the phase statistic is the *least* sensitive of the three (Table 4).

3 Operation order decides validity

3.1 The defect

The published pipeline applies OLS detrending, AR(1) prewhitening and (data dependent) rolling-volatility standardization to raw amplitudes, and ranks only afterwards. The ranks of the residual then depend on the unknown marginal F , so the calibration cannot be exact. We measured it: calibrate under a Gaussian null with $B = 1000$ replicates at $n = 10^4$, then test i.i.d. draws from six continuous marginals.

Table 1: Empirical size at nominal $\alpha = 0.05$ of the screening stage, calibrated under a Gaussian null. “KS” tests the p -values against $U(0, 1)$; 200 series per cell.

marginal	size@.05	size@.10	KS	p_{KS}
Gaussian	0.045	0.090	0.047	0.757
uniform	0.045	0.080	0.081	0.134
Student t_3	0.065	0.080	0.078	0.162
lognormal	0.190	0.245	0.237	< 0.001
exponential	0.065	0.110	0.107	0.018
Cauchy	0.880	0.885	0.840	< 0.001

At nominal 5% the test rejects i.i.d. Cauchy data 88% of the time. The cause is straightforward: for a marginal without finite moments the fitted trend and the AR(1) coefficient are unstable, and their instability writes marginal-dependent structure into the ranks of the residual.

3.2 The remedy, and its proof

Proposition 1 (distribution-freeness). *Let $x_1, \dots, x_n$ be i.i.d. from any continuous distribution F , and let T be any statistic that depends on the sample only through the vector of ranks. Then the distribution of T does not depend on F .*

Proof. Continuity gives distinct values almost surely, and the rank vector of an i.i.d. sample is uniform on the $n!$ permutations for every continuous F . A statistic measurable with respect to the rank vector therefore has a distribution determined by n alone. $\square$

The consequence is that the Monte-Carlo null is exact only if *every* amplitude operation is removed or performed after ranking. We compared four orders empirically (Table 2).

Table 2: Worst deviation of empirical size from the nominal 0.05 across the six marginals of Table 1, for four orders of operations. This is a separate, smaller run (150 series per cell, $B = 400$), so variant A reproduces the defect at 0.853 rather than the 0.880 of Table 1.

variant	description	worst size − 0.05
A	amplitude operations, then ranks (as published)	0.803
B	ranks, then the same amplitude operations	0.043
C	ranks only, no amplitude operations	0.030
D	ranks, then differencing	0.030

Only variant C passes the uniformity test at every marginal. Its cost is that trends and linear autocorrelation are no longer removed before screening, so the surrogate stage must carry that load alone; Section 5 shows it does so imperfectly (AR(1) with $\phi = 0.9$ becomes a false positive for the pair statistic). We regard the trade as clear-cut nonetheless: a test whose size is 0.85 on heavy tails is not usable, whereas a test that loses one hard case is.

The preprocessing was in any case partly redundant. Removing a linear trend and linear autocorrelation is exactly what the surrogate stage tests for; performing it twice, once destructively, gains nothing.

4 The procedure we recommend

1. **Symbolize.** Rank-normalize to $[0, 1]$ with average ranks for ties. No detrending, no prewhitening, no volatility scaling.
2. **Screen.** Compute the statistic on a grid (lattice resolutions $K \in \{100, 200, 400, 800\}$ for occupancy; pattern lengths d with $d!$ comparable to n for ordinal patterns). Standardize each cell by a Monte-Carlo null and take the maximum; obtain $p = (1 + \#\{T_{\text{null}} \geq T_{\text{obs}}\})/(1 + B)$. The multiplicity of the scan is inside the simulated null, so no Bonferroni correction is applied.
3. **Confirm.** If screening rejects, test against IAAFT surrogates [11], which reproduce the periodogram and the marginal to high but not exact numerical accuracy and therefore probe temporal ordering.
4. **Classify.** Confirmed structure that is extremely narrowband (spectral concentration > 0.95 or $|\text{ACF}| > 0.99$) is reported as *(quasi-)periodic*; otherwise as *confirmed nonlinear*, subdivided by transition occupancy into *low-occupancy*, *compatible with low-dimensional determinism* and *high-occupancy*, *nonlinear stochastic*. We deliberately avoid the label “chaos”: no surrogate test can establish determinism (Section 6).

Choosing the pattern length is not optional. For $n = 10^4$ the expected fraction of unobserved ordinal patterns under an i.i.d. null is $(1 - 1/d!)^{n-d+1}$: 0.0000 for $d \leq 6$, 0.138 for $d = 7$, 0.781 for $d = 8$, 0.973 for $d = 9$. At $d \leq 6$ the statistic is identically zero under the null, its variance vanishes, and any standardized score is meaningless. We report this because an earlier draft of this paper used $d \in \{4, 5, 6\}$ at $n = 10^4$ and obtained an apparently excellent baseline from a degenerate statistic. The trap is easy to fall into and, so far as we can see, is not flagged in the tooling.

5 Benchmark and results

5.1 The benchmark

165 series with provenance; 46 carry ground-truth labels. Synthetic labels are *computed*: maps by direct variational iteration, flows by Benettin renormalization on the ODE with the analytic Jacobian [12], giving $\lambda = 0.900$ (Lorenz, $\rho = 28$), 0.083 (Rössler, $c = 5.7$), 0.425 (Chua) and negative values for the regular regimes. Stochastic and mixed cases are labeled by construction, not by an exponent — a distinction we now state explicitly, since “detector-independent ground truth” overstated it.

Computing λ corrected a label carried in the v1 manifest, and the correction is delicate. At $r = 3.57$ the logistic map has $\lambda = 0.01096$ with a bootstrap CI of $[0.0109, 0.0111]$, so it is weakly chaotic rather than “Regular”. But $r = 3.57$ lies *above* the Feigenbaum accumulation point $r_\infty = 3.5699457$ (where $\lambda \approx 2 \times 10^{-5}$), and at $r = 3.5705$ one finds $\lambda = -0.0039$: the parameter sits

in a region riddled with periodic windows. We report the case as correctly labeled but hypersensitive, and all three statistics miss it.

The 46 labeled series are not 46 independent systems: several are coordinates of one attractor or members of a parameter family. Accuracy figures should be read accordingly, and we report class-wise counts rather than a single number where possible.

5.2 Matched comparison

All three statistics were run through the identical protocol of Section 4 — same calibration, same surrogate stage, same decision rule, same auxiliary criteria. Only the discriminating statistic differs. Scoring is strict: the six *Mixed* (chaos+noise) cases are excluded rather than counted as correct under either label, a rule inherited from v1 that inflated the apparent score by six unloseable cases.

Table 3: Matched comparison, preprocessing-free, 40 labeled series, strict scoring. Size is the range of empirical size at nominal 0.05 over the six marginals of Table 1.

statistic	accuracy	empirical size range
missing ordinal patterns, $d \in \{6, 7, 8\}$	37/40 = 92.5%	0.067–0.100
pair occupancy ($N_t, \Delta N_t$)	35/40 = 87.5%	0.033–0.092
cyclotomic phase scan	34/40 = 85.0%	0.025–0.067

Table 4: Class-wise counts for the same runs. The phase statistic is the most conservative and the least sensitive; the ordinal statistic is mildly liberal and the most sensitive.

	ordinal	occupancy	phases
Chaos ($n = 14$)	13/14	12/14	9/14
Regular ($n = 10$)	9/10	9/10	9/10
Noise ($n = 16$)	15/16	14/16	16/16

Three cases are missed by every statistic: the logistic map at $r = 3.57$, a sine at SNR ≈ 10 dB whose label is debatable, and GARCH volatility (Section 6). The ordering in Table 3 is monotone in the amount of machinery: the classical baseline is best, the plain occupancy count second, the cyclotomic construction last. A gap of two cases in forty is not statistically decisive, and we do not claim the ordinal statistic is proven superior; we claim the phase construction is not shown to be superior, which given its cost is sufficient.

6 Measured limits

6.1 The surrogate stage leaks

Every verdict rests on the surrogate stage, so its size must be measured rather than assumed. Table 5 gives rejection rates at nominal 0.05.

A static monotone transform of a linear Gaussian process is inside the null by construction, and is rejected in a quarter of replicates — a known weakness of amplitude-adjusted surrogates [13, 14], here quantified for this class of statistics. Heteroscedastic processes are rejected at 50–75%, which is the decisive number for applications.

Table 5: Rejection rate of the IAAFT confirmation stage. The first block lies inside the IAAFT null, where any rejection is a false positive. 8 replicates per process.

process	screening < .05	IAAFT < .05
white Gaussian; AR(1) $\phi = 0.5, 0.9$	0.00	0.00
sine + noise; i.i.d. t_3	1.00 / 0.12	0.00
AR(2) oscillatory; ARMA(2,2); long memory	1.00	0.12
cubed AR(1) (static monotone transform)	1.00	0.25
GARCH(1,1) returns	0.50	0.50
stochastic volatility	0.75	0.75

6.2 Consequences for the applied claims

The financial conclusion is withdrawn. Both v1 and our earlier draft read S&P 500 log returns as carrying nonlinear structure. Given Table 5, that reading is unsupported: heteroscedastic series are rejected by this stage at 50–75% regardless of nonlinear dynamics, and returns are heavy-tailed, where Table 1 shows the published pipeline was also miscalibrated. Testing financial returns requires a model-based null (constrained GARCH surrogates), which we have not run. The correct present statement is that the test is uninformative for this class.

The climate conclusion is softened. Daily HadCET anomalies, monthly Niño3.4, PDO and AMO show no evidence of low-dimensional determinism *within this test*, consistent with a stochastic climate model [15] and with the resolution of the 1980s attractor controversy [16, 17]. Absence of detection is not evidence of absence. The methodological point stands independently: screening alone flags most climate series as “structured”, so a pipeline without the surrogate stage would reproduce the error of that era.

The EEG gradient is confounded. On the Bonn data [18] the confirmed-nonlinear rate rises monotonically across classes (1/20, 0/20, 1/20, 4/20, 17/20 from healthy to ictal). But classes A/B are scalp recordings and C/D/E are intracranial, differing in modality and amplitude resolution; the gradient tracks that as well as clinical severity. No clinical claim is made.

6.3 Confounds inherent to the approach

Nonlinearity is not determinism. GARCH volatility is nonlinear (the surrogate stage correctly rejects) and smooth (transition occupancy 0.41, inside the deterministic range), so it is misclassified. A random walk sits at 0.17. This is the effect described in [8, 9] and it is a limit of surrogate methodology, not a threshold to be tuned.

Instrumental quantization manufactures the signal. RR intervals at 128 Hz carry 50–79 distinct values in 10^4 points; coarse digitization creates transition sparsity for free. Referencing the occupancy statistic against a random permutation of the *same values* cancels the value multiset exactly and largely removes the artifact (0.04–0.09 before, 0.26–0.63 after). We report a low-resolution flag and decline to interpret the HRV verdicts physiologically.

Densely sampled flows. Most structure of a smooth oversampled flow lives in the spectrum, which IAAFT reproduces, so the surrogate stage has little power. Reducing the flow to successive local maxima repairs this on our six flows, including the two that fail on the raw series. We flag

this as an in-sample result: the reduction was devised on the cases that motivated it and has not been validated out of sample.

7 What is new, what is not, and what remains

Not new. The detection principle: transition occupancy and forbidden/missing patterns in symbolic dynamics [1, 2, 3, 5, 4]. The mathematics: the cyclotomic factorization, the order of an element, and the primitive-divisor congruence are classical [10]. The smooth-stochastic confound [8, 9]. Surrogate methodology [19, 11].

New, as far as we can determine. (i) The cyclotomic identification of the phase lattice, Eq. (1) with Lemma 1 — correct, and negative in consequence. (ii) The demonstration that operation order determines distribution-freeness for this family of tests, with measured sizes up to 0.85 and a remedy (Proposition 1, Table 2). (iii) Measured empirical sizes for three symbolic statistics across six marginals, and a size audit of the IAAFT stage for this class (Table 5). (iv) The saturation trap in specifying the ordinal baseline at fixed n . (v) An open benchmark with computed rather than asserted synthetic labels.

Useful, and missing from current tooling. Widely used packages supply these statistics but not the inference around them: `ordpy` [20] computes permutation entropy, complexity–entropy planes, missing patterns and ordinal networks, and provides no hypothesis test, no p -values, no surrogate framework and no null calibration; surrogate generators exist separately and are not wired to the statistics. The ordinal literature itself notes that raw missing-pattern counts cannot be thresholded by inspection, since they arise from correlated stochastic processes as well as from determinism [4]. The gap is a decision layer, and that is what this paper supplies and validates.

Not done. Power curves over n and SNR at fixed size, instead of accuracy at a single α ; a frozen protocol evaluated on a held-out set of generators (the thresholds and the kernel choice here were selected on the same 46 series); model-based surrogates for heteroscedastic data; out-of-sample validation of the return-map reduction; several realizations per system with cluster-aware uncertainty. These are the experiments we would require of this paper as referees, and we flag their absence rather than work around it.

8 Conclusion

The construction proposed in v1 does not survive: it is a lossy projection of a pair statistic that is itself a rediscovery of a classical mechanism, and at matched nominal size it is the weakest of the three statistics we compared. The cubic kernel has an exact cyclotomic explanation, which is satisfying and useless.

What survives is a procedure and three measurements. Rank-only symbolization makes the test exactly distribution-free, which the published pipeline was not; the surrogate stage that carries every verdict leaks at rates we now quantify; and the choice of pattern length silently destroys the standard baseline at common sample sizes. Together with an open benchmark whose synthetic labels are computed, these make a working, honestly bounded screening procedure — which is less than v1 claimed, and more than it delivered.

Reproducibility

Code, benchmark manifest and results are in the v2 branch of the project repository, <https://github.com/Theclimateguy/DDBM/tree/v2>: `build_dataset.py` regenerates or downloads all series; `rank_only_pair.py` and `ordinal_scaled.py` reproduce Tables 3 and 4; `pivotality_fix.py` reproduces Table 2; `reviewer_experiments.py` reproduces Tables 1 and 5 and the Lyapunov intervals; `cyclotomic_kernels.py` the symbolic derivations. Seeds are fixed; series files are not redistributed and are rebuilt from primary sources.

Acknowledgements

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